In [1]:
```python
a = "Sam"
b = 8
c = 9807654999
d = "Bangalore"
```

In [2]:
```python
a
```

Out[2]: 'Sam'

In [3]:
```python
tiger = ["Sam",8,9807654999,"Bangalore"]
```

In [4]:
```python
tiger[0]
```

Out[4]: 'Sam'

In [5]:
```python
tiger [2]
```

Out[5]: 9807654999

In [7]:
```python
simba = ["Chandu",10,8374699024,"Hyderabad"]
```

In [8]:
```python
simba
```

Out[8]: ['Chandu', 10, 8374699024, 'Hyderabad']

In [10]:
```python
simba [3]
```

Out[10]: 'Hyderabad'

In [11]:
```python
star = ["Nak",5,9645876212,"Hyderabad"]
```

In [12]:
```python
star
```

Out[12]: ['Nak', 5, 9645876212, 'Hyderabad']

In [13]:
```python
tiger.insert(2,"Bangalore")
```

In [22]:
```python
tiger
```

Out[22]: ['Sam', 8, 'Bangalore', 9807654999]

In [15]:
```
simba.insert(2,"Paris")
```

In [21]:
```
simba
```

Out[21]:
```
['Chandu', 10, 'Paris', 8374699024]
```

In [17]:
```
star.insert(2,"Hyderabad")
```

In [20]:
```
star
```

Out[20]:
```
['Nak', 5, 'Hyderabad', 9645876212]
```

In [32]:
```
tiger.pop()
simba.pop()
star.pop()
```

Out[32]:
```
'Data Tech Accelerator'
```

In [23]:
```
tiger.append("sam@google.com")
simba.append("chandu@loreal.com")
star.append("nak@microsoft.com")
```

In [33]:
```
print(tiger,simba,star)
```

```
['Sam', 8, 'Google', 'Bangalore', 9807654999, 'sam@google.com', 'Cloud D
ata Archetict'] ['Chandu', 10, 'Loreal', 'Paris', 8374699024, 'chandu@lo
real.com', 'Data Tech Accelerator'] ['Nak', 5, 'Microsoft', 'Hyderabad',
9645876212, 'nak@microsoft.com', 'Global Fabric Head']
```

In [27]:
```
tiger.insert(2,"Google")
simba.insert(2,"Loreal")
star.insert(2,"Microsoft")
```

In [30]:
```
tiger.append("Cloud Data Archetict")
simba.append("Data Tech Accelerator")
star.append("Global Fabric Head")
```

In [106]:
```
lst = [2,5,7,2,3,9,12,20,15,12,9]
```

In [44]:
```
lst.sort()
```

In [62]:
```
lst
```
Out[62]:
```
[2, 5, 7, 2, 3, 9, 12, 20, 15, 12, 9]
```
In [107]:
```
lst.insert(6,77)
```
In [64]:
```
lst
```
Out[64]:
```
[2, 5, 7, 2, 3, 9, 77, 12, 20, 15, 12, 9]
```
In [65]:
```
lst.sort()
```
In [66]:
```
lst
```
Out[66]:
```
[2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```
In [42]:
```
lst.clear()
```
In [57]:
```
lst.remove(9) #we have to give values to remove in the list
```
In [67]:
```
lst
```
Out[67]:
```
[2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```
In [68]:
```
list1 = lst.copy()
```
In [69]:
```
list1
```
Out[69]:
```
[2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```
In [59]:
```
lst.remove(12)
```
In [70]:
```
lst
```
Out[70]:
```
[2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```

In [ ]:
```
#how to remove the 2nd element?         Answer: This is directly not possib
#how to sort in descending order?       Answer: This is possible based what
#how to remove duplicates at once?      Answer: This is directly not possib
#how to remove values based on index?   Answer: This is possible based what
```

In [ ]:
```

```

In [71]:
```
a =[1,2,3]
b =[7,7,7]
```

In [72]:
```
a.extend(b)
```

In [73]:
```
a
```

Out[73]:
```
[1, 2, 3, 7, 7, 7]
```

In [78]:
```
lst.index(2)
```

Out[78]:
```
0
```

In [85]:
```
lst
```

Out[85]:
```
[2, 5, 7, 2, 3, 9, 12, 20, 15, 12, 9]
```

In [80]:
```
lst.remove(2)
```

In [81]:
```
lst
```

Out[81]:
```
[2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```

In [82]:
```
lst.remove(2)
```

In [92]:
```
lst
```

Out[92]:
```
[2, 5, 7, 2, 3, 9, 77, 12, 20, 15, 12, 9]
```

In [109]:
```
lst.sort()
```

In [110]:
```
lst
```

Out[110]:
```
[2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]
```

In [ ]:
```
#how to sort in descending order?
```

In [98]:
```
lst.sort(reverse=True)
```

In [99]:
```
lst
```

Out[99]:
```
[77, 20, 15, 12, 12, 9, 9, 7, 5, 3, 2, 2]
```

In [ ]:
```
##how to remove the 2nd element?
```

In [111]:
```
first_index = lst.index(9)
```

In [112]:
```
first_index
```

Out[112]:
```
5
```

In [113]:
```
second_index = lst.index(9,first_index+1)
```

In [114]:
```
second_index
```

Out[114]:
```
6
```

In [115]:
```
lst.pop(second_index)
```

Out[115]:
```
9
```

In [116]:
```
lst
```

Out[116]:
```
[2, 2, 3, 5, 7, 9, 12, 12, 15, 20, 77]
```

In [ ]:
```
##how to remove duplicates at once?

lst = [2, 2, 3, 5, 7, 9, 9, 12, 12, 15, 20, 77]

# Elements to remove
elements_to_remove = {2, 9}

# Remove all occurrences of elements in elements_to_remove
lst = [x for x in lst if x not in elements_to_remove]

print(lst)  # Output: [3, 5, 7, 12, 12, 15, 20, 77]
```

In [ ]:
```
##how to remove values based on index?
```

In [117]:
```
lst
```

Out[117]: [2, 2, 3, 5, 7, 9, 12, 12, 15, 20, 77]

In [118]:
```
lst.pop(1)
```

Out[118]: 2

In [119]:
```
lst
```

Out[119]: [2, 3, 5, 7, 9, 12, 12, 15, 20, 77]

In [121]:
```
lst.pop(5)
```

Out[121]: 12

In [122]:
```
lst
```

Out[122]: [2, 3, 5, 7, 9, 12, 15, 20, 77]

In [ ]:
```
#dictionary
```

In [1]:
```
sam = ("Venkat",60,"Farmer",9959927644,"Mallavaram","AndhraPradesh")
```

In [2]:
```
type(sam)
```

Out[2]: tuple

In [11]:
```
sam.index(9959927644) #index will find the occurence of element
```

Out[11]: 3

In [12]:
```
sam.count(60)
```

Out[12]: 1

In [13]:
```
sam1 = list(sam)
```

In [14]:
```
sam1
```
Out[14]:
```
['Venkat', 60, 'Farmer', 9959927644, 'Mallavaram', 'AndhraPradesh']
```
In [15]:
```
sam1.append("India")
```
In [16]:
```
sam1
```
Out[16]:
```
['Venkat', 60, 'Farmer', 9959927644, 'Mallavaram', 'AndhraPradesh', 'Ind
ia']
```
In [17]:
```
sam = tuple(sam1)
```
In [18]:
```
sam
```
Out[18]:
```
('Venkat', 60, 'Farmer', 9959927644, 'Mallavaram', 'AndhraPradesh', 'Ind
ia')
```
In [20]:
```
v = {"Name" : "Venkat" , "Age" : 60, "Occupation" : "Farmer", "Ph no": 995
```
In [21]:
```
type(v)
```
Out[21]:
```
dict
```
In [22]:
```
v
```
Out[22]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'AndhraPardesh'}
```
In [23]:
```
v.copy()        #copying the content
```
Out[23]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'AndhraPardesh'}
```
In [31]:
```
v.update(Address = "Mallavaram,Andhra Pradesh") #updating the existing rec
```

In [33]:
```
v.values() #display the values only
```

Out[33]:
```
dict_values(['Venkat', 60, 'Farmer', 9959927644, 'Mallavaram,Andhra Prad
esh'])
```

In [34]:
```
v.keys() #display the keys
```

Out[34]:
```
dict_keys(['Name', 'Age', 'Occupation', 'Ph no', 'Address'])
```

In [37]:
```
v.get("Name") # getting the name key value
```

Out[37]:
```
'Venkat'
```

In [38]:
```
v.update({"Vehicle": "TVS Radeon"})
```

In [39]:
```
v
```

Out[39]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon'}
```

In [40]:
```
v.update({"House" : "2BHK"})
```

In [41]:
```
v
```

Out[41]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon',
 'House': '2BHK'}
```

In [42]:
```
v.pop("House")
```

Out[42]:
```
'2BHK'
```

In [43]:
```
v
```

Out[43]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon'}
```

In [44]:
```python
v.items()
```

Out[44]:
```
dict_items([('Name', 'Venkat'), ('Age', 60), ('Occupation', 'Farmer'),
('Ph no', 9959927644), ('Address', 'Mallavaram,Andhra Pradesh'), ('Vehic
le', 'TVS Radeon')])
```

In [45]:
```python
#Assignment
# is it possible to update dict's key?
# can I add key/value pair in between elements?
```

In [50]:
```python
v.setdefault("Age")
```

Out[50]:
```
60
```

In [51]:
```python
v
```

Out[51]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon'}
```

In [52]:
```python
v.popitem()
```

Out[52]:
```
('Vehicle', 'TVS Radeon')
```

In [53]:
```python
v.update({"Vehicle": "TVS Radeon"})
```

In [54]:
```python
v
```

Out[54]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon'}
```

In [55]:
```python
v1 = list(v)
```

In [56]:
```python
v1
```

Out[56]:
```
['Name', 'Age', 'Occupation', 'Ph no', 'Address', 'Vehicle']
```

In [58]:
```
v1[0] = "FullName"
```

In [59]:
```
v1
```

Out[59]:
```
['FullName', 'Age', 'Occupation', 'Ph no', 'Address', 'Vehicle']
```

In [61]:
```
v
```

Out[61]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon'}
```

In [63]:
```
v.update({"FullName" : "Ram"})
```

In [64]:
```
v
```

Out[64]:
```
{'Name': 'Venkat',
 'Age': 60,
 'Occupation': 'Farmer',
 'Ph no': 9959927644,
 'Address': 'Mallavaram,Andhra Pradesh',
 'Vehicle': 'TVS Radeon',
 'FullName': 'Ram'}
```

In [65]:
```
v.popitem()
```

Out[65]:
```
('FullName', 'Ram')
```

In [69]:
```
## is it possible to update dict's key?
#It is directly not possible based learned topics sofar,
#But If we create another dictionary with updating the each and very eleme
```

In [66]:
```
keys_update = {
    'Full Name' : v['Name'],
    'Age in years' : v['Age'],
    'Job' : v['Occupation'],
    'Phone number' : v['Ph no'],
    'Location' : v['Address'],
    'Vehicle Mode': v['Vehicle']
}
```

In [67]:
```
keys_update
```

Out[67]:
```
{'Full Name': 'Venkat',
 'Age in years': 60,
 'Job': 'Farmer',
 'Phone number': 9959927644,
 'Location': 'Mallavaram,Andhra Pradesh',
 'Vehicle Mode': 'TVS Radeon'}
```

In [73]:
```
v.insert({'Name','Price': "55lakhs"})
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[73], line 1
----> 1 v.insert({'Price': "55lakhs"})

AttributeError: 'dict' object has no attribute 'insert'
```

In [77]:
```
# can I add key/value pair in between elements? A: It is not possible. ask
```

Out[77]:
```
{'A': None, 'g': None, 'e': None}
```